

WAKALAH

The trust layer for AI-agent transactions.

AI agents book, buy, and send money on people's behalf.

Nobody can answer: is this agent still acting for a real, present, un-hijacked human? Wakalah answers with the mobile network's own records.

Zaid Takieddin + Yasser Al-Koudmany | Live demo + working system, not slides

Credentials are not identity.

- > **Agents spend money now.** Booking, buying, and remitting on a human's behalf - every week, more of it.
- > **Every scheme checks the agent.** APIs, tokens, OAuth scopes: all of them authorize credentials. None check the human behind them.
- > **Takeover looks identical.** Stolen credentials plus a cloned agent are indistinguishable at the protocol layer.
- > **MENA pays first.** Remittance corridors run on SIM-based identity, and every fraud playbook here starts with a SIM swap - soon at machine speed.

A mandate, a verification, a kill-switch.

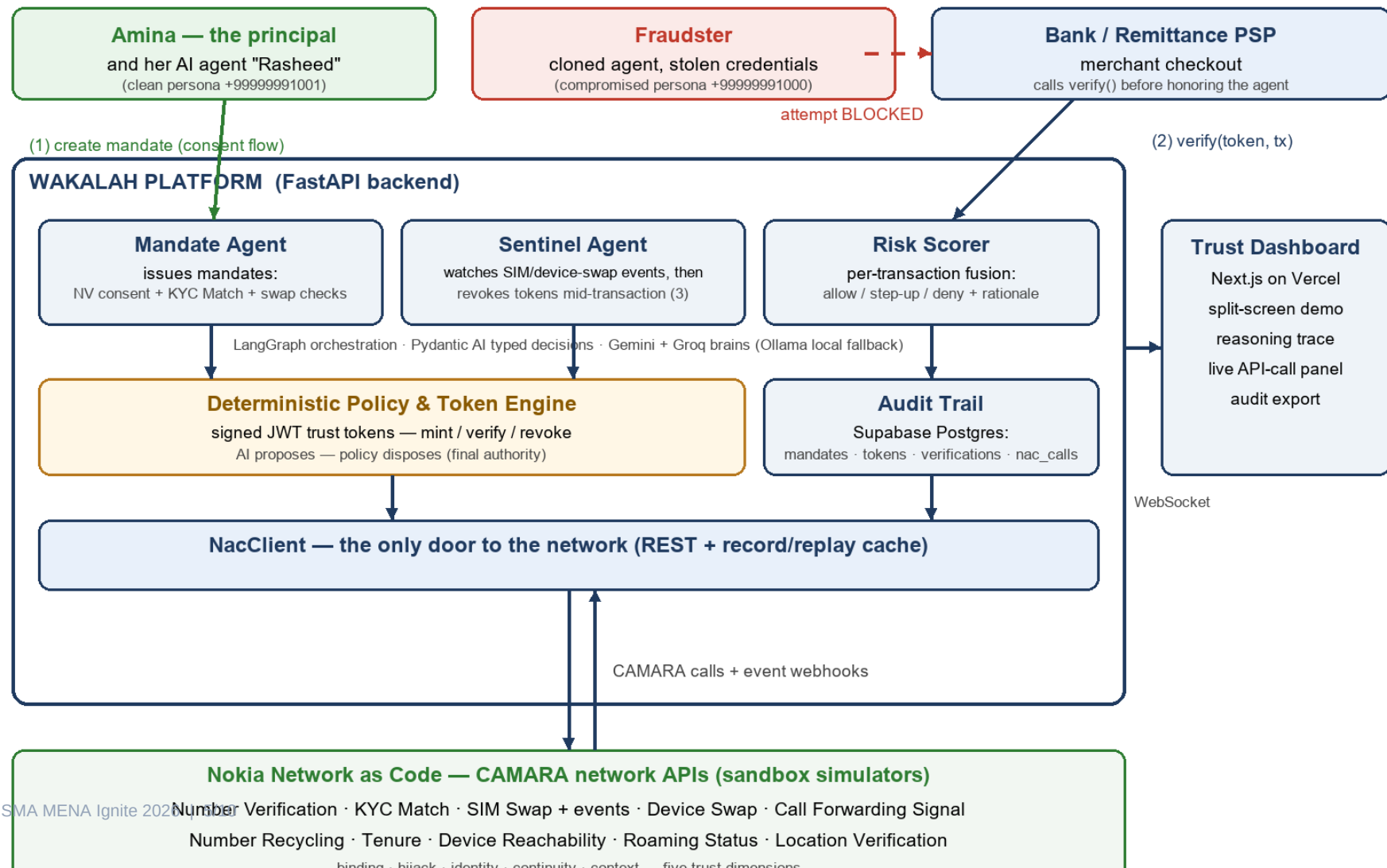
- > **MANDATE.** Amina authorizes her agent once: scoped, expiring, bound to its key. Paid to her mother, up to 2,000/month, nothing else.
- > **VERIFY.** Every transaction is risk-classified by AI, checked against live network signals, and decided by a deterministic policy engine.
- > **REVOKE.** The moment the operator reports a SIM swap, every mandate dies - mid-transaction if necessary. No polling, no grace period.
- > **AI proposes, policy disposes.** The agent may add friction. It can never remove it, and it can never refuse. Every override is shown on screen.

Watch the system work harder when it matters.

ALLOW	Routine 150 to mother: 1 check, approved.
CHALLENGE	1,500 first payment to landlord: 4+ checks, stepped up.
DENY	Unknown payee: outside the mandate, refused with no network call.
REVOKE	Operator pushes a SIM swap: mandate dies in seconds.
DENY	Cloned agent demands 1,800: SIM swap + device swap + forwarding + recycled.
CHALLENGE	Operator outage: missing evidence is never approval.

Live: <https://wakalah-balance-demo.vercel.app/scenario> | Same beats, same verdicts, every run.

Four layers, one trust decision.



Ten signals. Five questions. Zero guesses.

- > **BINDING:** Number Verification - 3-legged consent flow, implemented end to end.
- > **IDENTITY:** KYC Match at transaction time, KYC Fill-in at onboarding.
- > **HIJACK:** SIM Swap (+ operator events), Device Swap, Call Forwarding Signal.
- > **CONTINUITY:** Number Recycling, Tenure - is this still the same human?
- > **CONTEXT:** Reachability, Roaming, Location Verification - is the situation consistent?

Reasoning you can watch. Authority you can audit.

- > **SUPERVISOR (LangGraph):** classify risk -> build a verification plan -> enforce the floor -> gather evidence -> interpret -> decide. Every step streams live.
- > **BRAINS:** Gemini Flash reasons, Flash-Lite takes the hot path, Ollama stands by offline. Typed proposals only - prose never decides.
- > **POLICY v2:** versioned YAML rules with per-tier minimum floors. First material payment to a new beneficiary always steps up.
- > **THE MOMENT:** agent says ALLOW, policy says CHALLENGE or DENY - shown side by side, every time. That split IS the product.

A demo judges can believe.

- > **HONESTY LABELS:** every signal carries live / cached / replay / simulated / unavailable. Never rendered as what it is not.
- > **DETERMINISTIC:** scripted 7-beat story plus a recorded-response fallback. Dead venue wifi changes nothing on screen.
- > **TYPED EVERYTHING:** every cross-layer event validated against a schema; ad-hoc shapes fail loudly instead of leaking.
- > **AUDITED:** Supabase trail keyed by hashed principal reference, degrading gracefully to memory. 129 offline tests, strict lint + type gates.

Systems, not screenshots.

129

offline tests
(~2 seconds)

7/7

demo beats land
as documented

10

network signals
5 dimensions

0

API keys needed
to run the demo

App: <https://wakalah-balance-demo.vercel.app> | API: <https://wakalah.onrender.com/health>

Repo: <https://github.com/ZaidTakieddin/wakalah> | Run it: README RUNNING.md (replay mode, zero keys)

The network becomes the identity layer.

- > **OPERATORS EARN:** every agent transaction is billable verification calls - fraud prevention as a revenue line, not a cost center.
- > **BANKS SAVE:** takeover stopped before authorization; travelers stop eating false declines (roaming explains location).
- > **MENA FIRST:** SIM-registered identity + remittance corridors = the region where this matters most, built by its developers.
- > **THE ASK:** a live demo slot and your hardest questions. Team Wakalah: Zaid Takieddin (backend, agents, NaC) + Yasser Al-Koudmany (frontend console). Thank you.